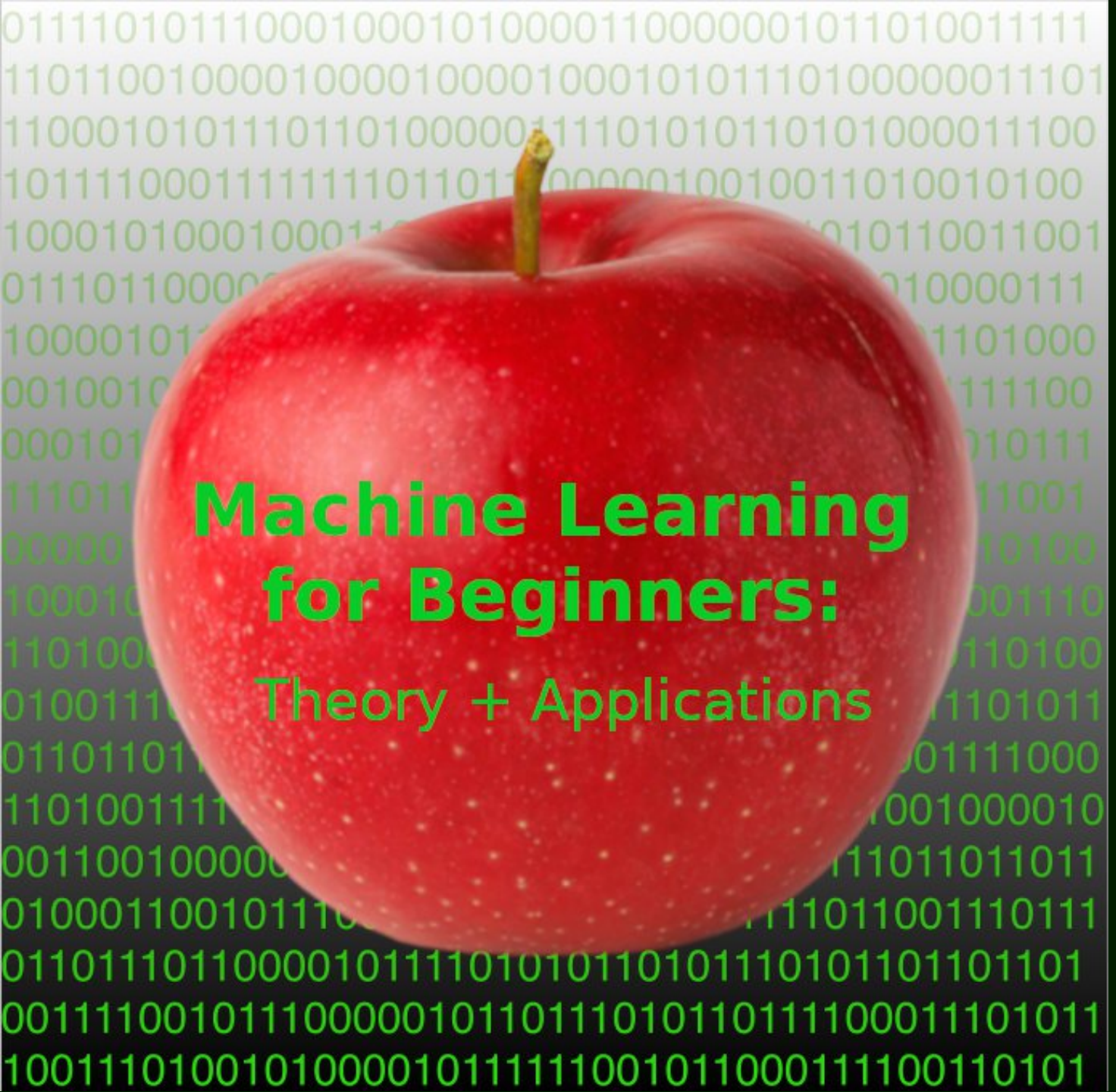


University of Bayreuth
Chair for Polymer Engineering
Prof. Dr. Holger Ruckdäschel

Title:

"Machine Learning for Beginners: Theory and Applications"

(Available in E-learning/Moodle)

Lecturer:

Dr. Rodrigo Q. Albuquerque

Lecture 9

Topics

Week 1: Introduction
ML types, data, ethics, case studies

Week 2: Preprocessing & *k*NN
Preprocessing, *k*-nearest neighbors, distance metrics, hyperparameters, cross-validation

Week 3: Linear Regression
Types, loss function, (over/under)fitting bias/variance, L1/L2 regularization, metrics

Week 4: Classification
Logistic regression, SVM, classification evaluation metrics, confusion matrix one-hot/label encoding

Week 5: Random forests
Decision trees, random forests, classification and regression applications

Week 6: PCA / KMeans
Dimensionality reduction, clustering

Week 7: Extra exercises

Week 8: Simple ANNs
Feedforward networks, activation functions, basic training

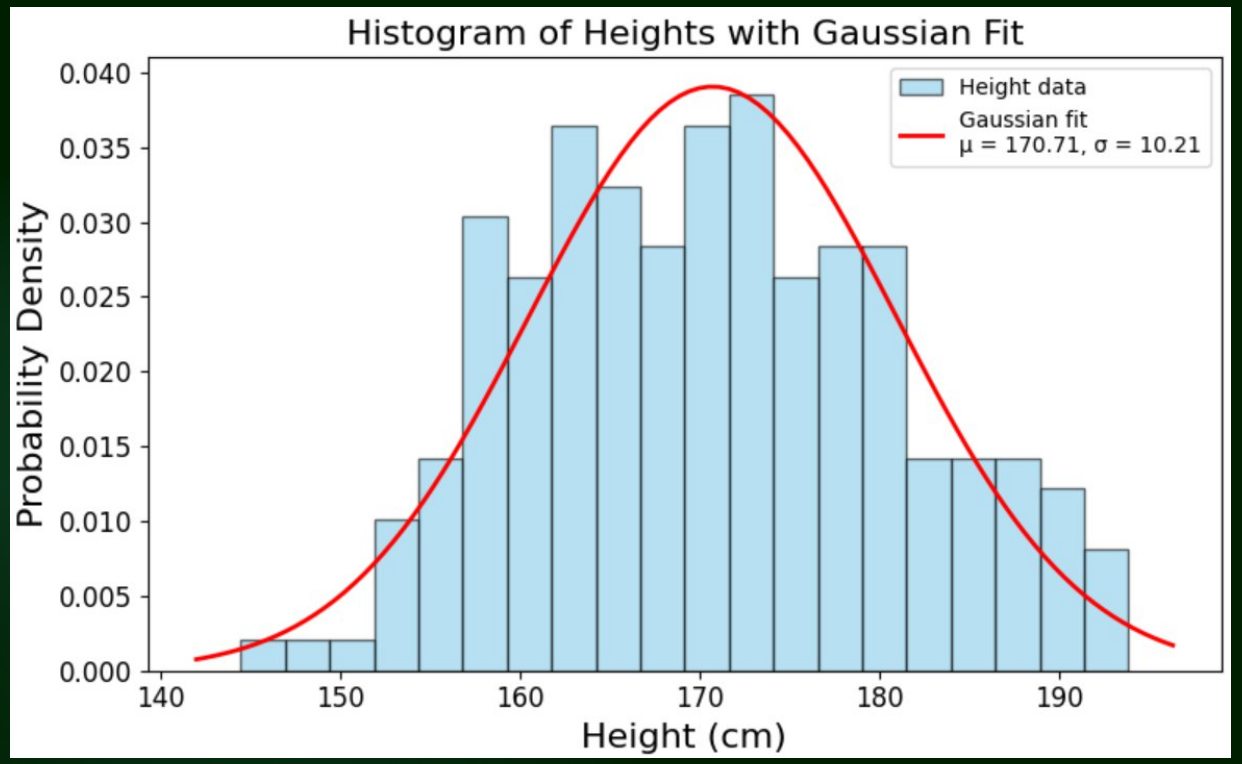
★ **Week 9: Gaussian Processes**
Kernels, gaussian processes

Week 10: Design of Experiments
DoE, random design, active learning, bayesian optimization

Weeks 11-12: Student projects

Gaussian distribution (1D)

How people's heights are distributed?



Gaussian (or "normal") distribution

$$p(x \mid \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

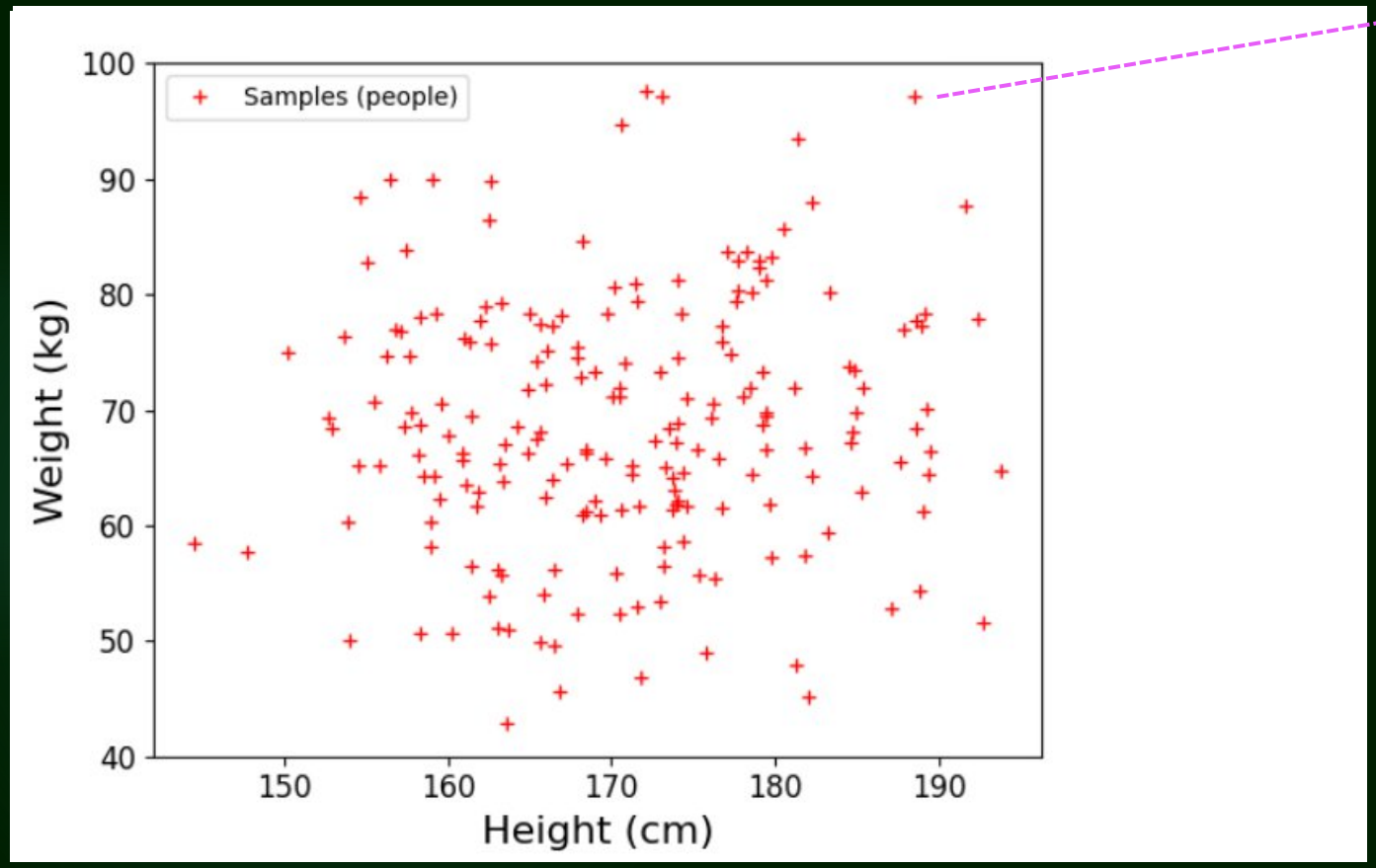
Probability density function (PDF)

or, we can say:

$$x \sim \mathcal{N}(\mu, \sigma^2)$$

Gaussian distribution (2D)

How people's heights and weights are distributed?



Sample vector: $x = [188\text{ cm}, 97\text{ kg}]$

Probability density function (PDF)

$$p(x \mid \mu, \Sigma) = \frac{1}{(2\pi)^{d/2}\sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right)$$

For 2 dimensions (d = 2). $x = [\text{height}, \text{weight}]$

p is a joint PDF: p(Height , Weight)

Fitted Gaussian parameters:

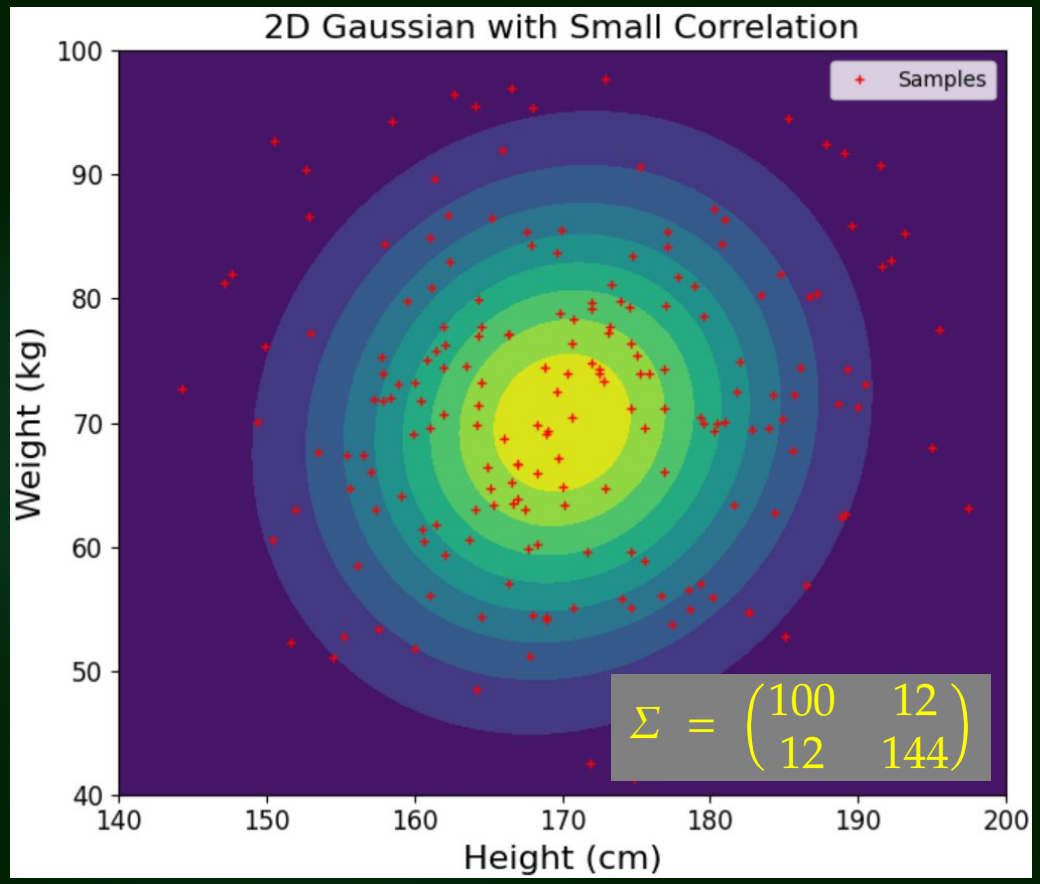
Mean vector: $\mu = [171, 68]$

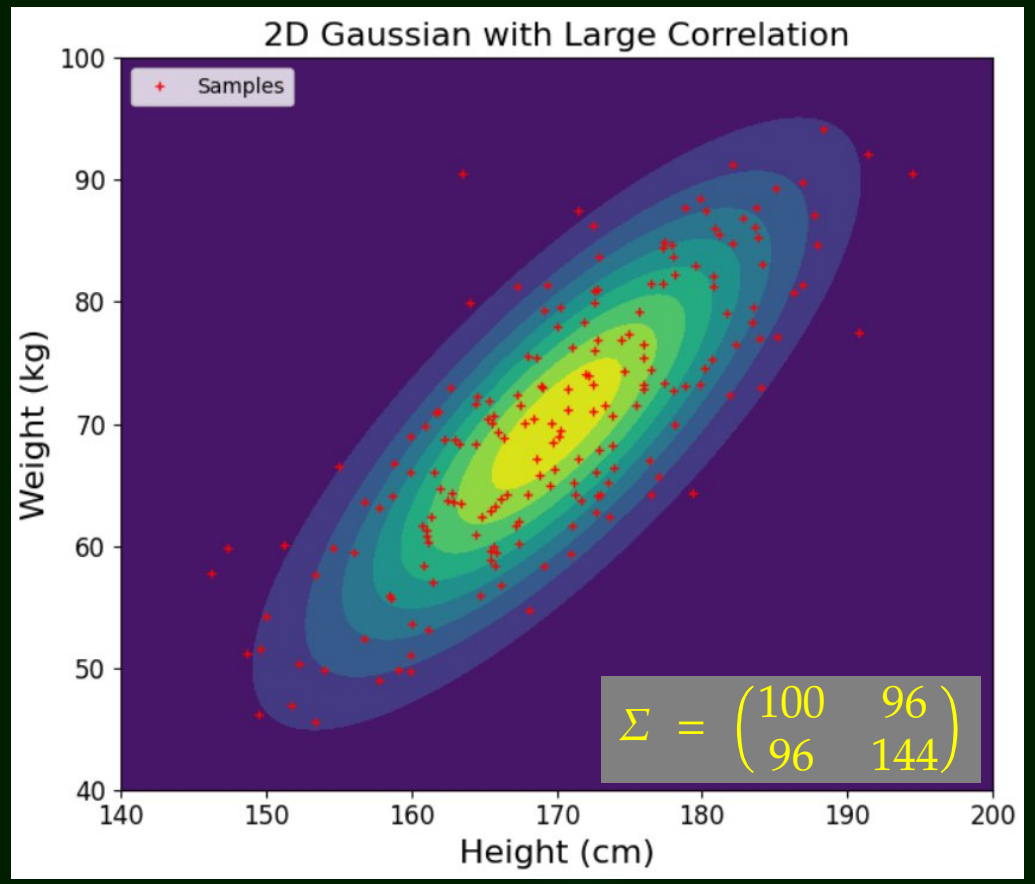
Covariance matrix: $\Sigma = \begin{pmatrix} 100 & 12 \\ 12 & 144 \end{pmatrix}$

We can use the fitted 2D Gaussian to draw new samples

Gaussian distribution (2D)

How correlated are the variables?

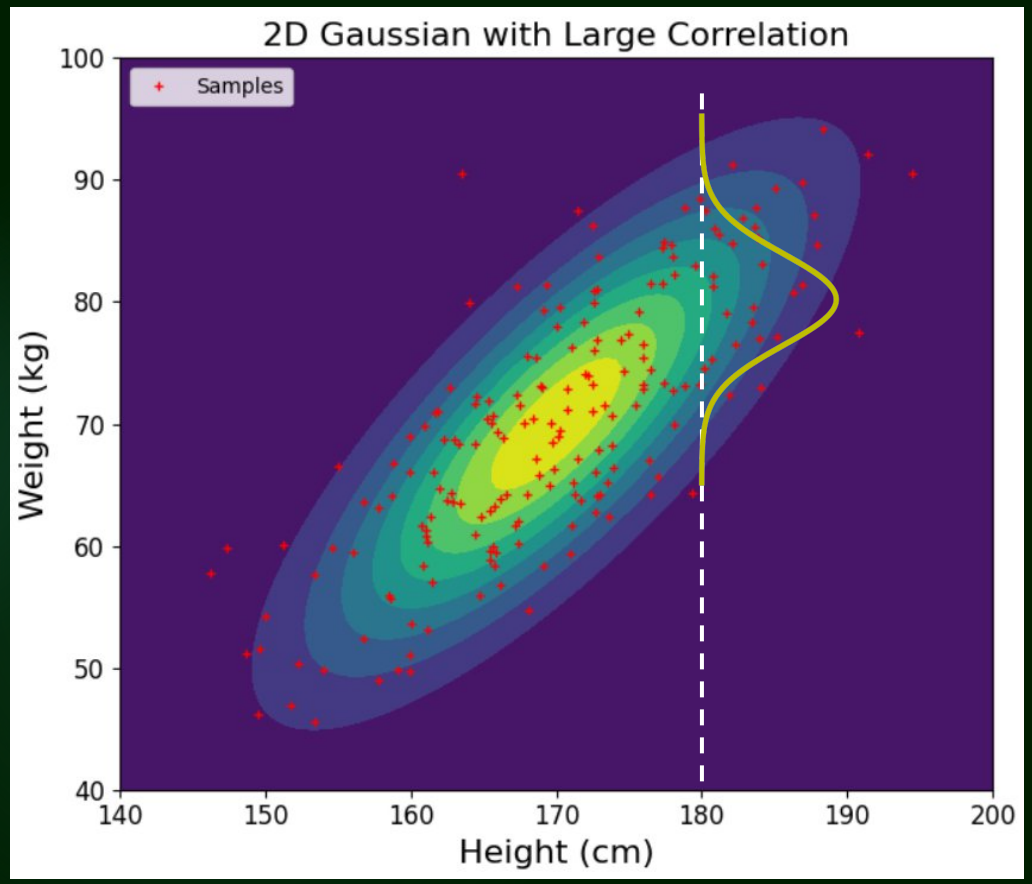




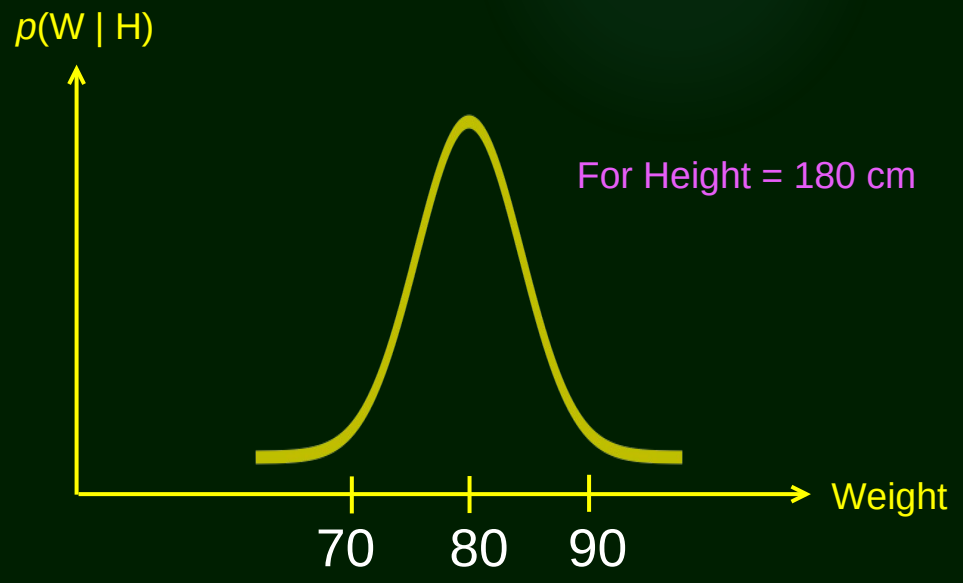
Note: If features are standardized, the main diagonal of Σ become 1's

Gaussian distribution (2D)

Conditional probability: $p(W \mid H = 180)$



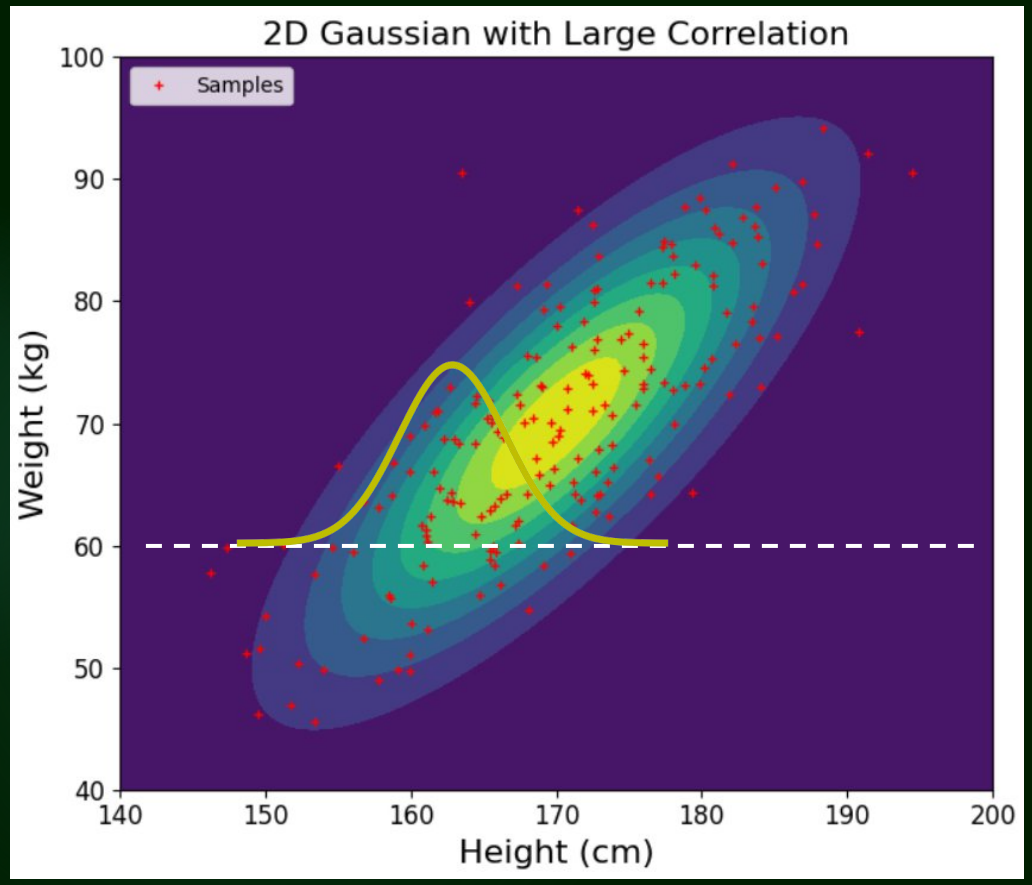
Question: If the height is e.g., 180 cm, what is the most probable weight?



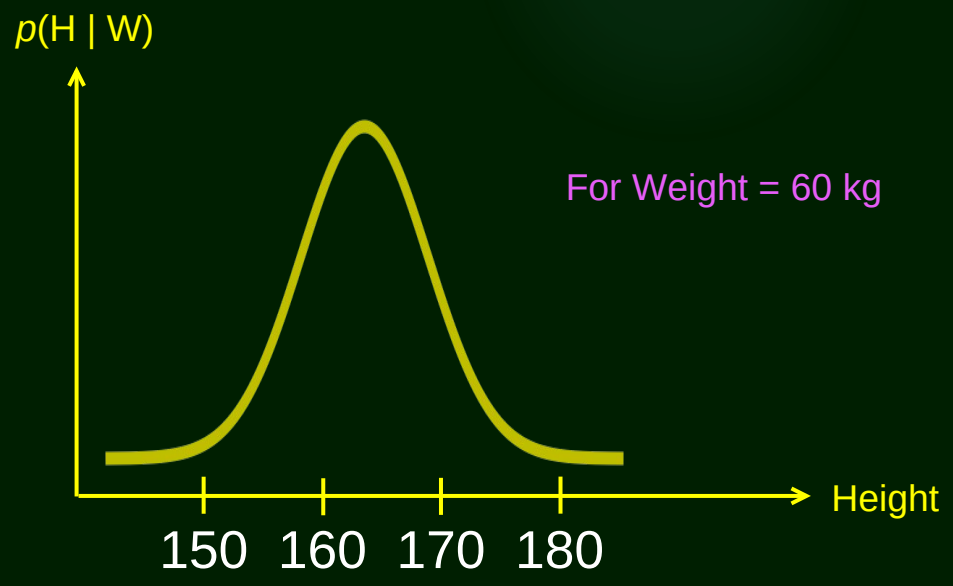
One gets $p(W \mid H)$ analytically from the original joint distribution $p(W , H)$

Gaussian distribution (2D)

Conditional probability: $p(\mathrm{H} \mid \mathrm{W} = 60)$



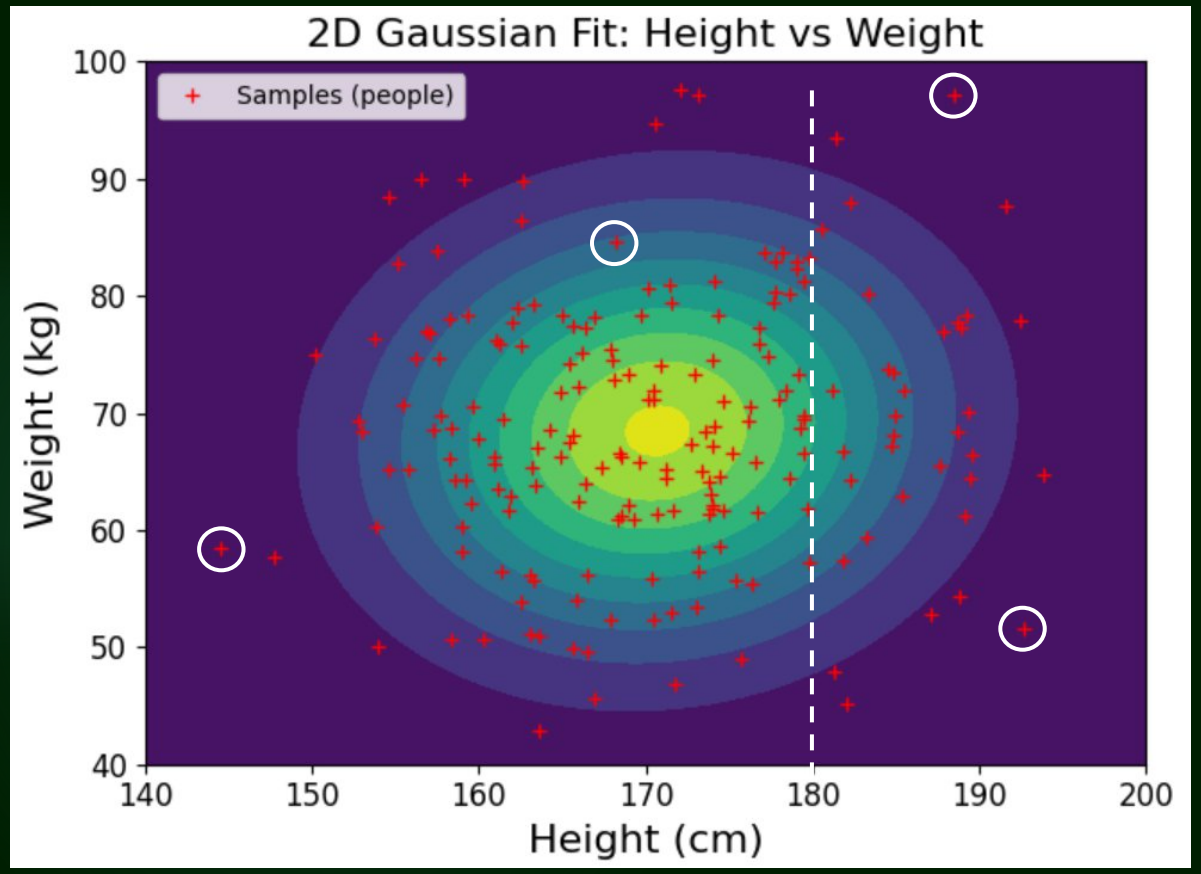
Question: If the weight is e.g., 60 kg, what is the most probable height?



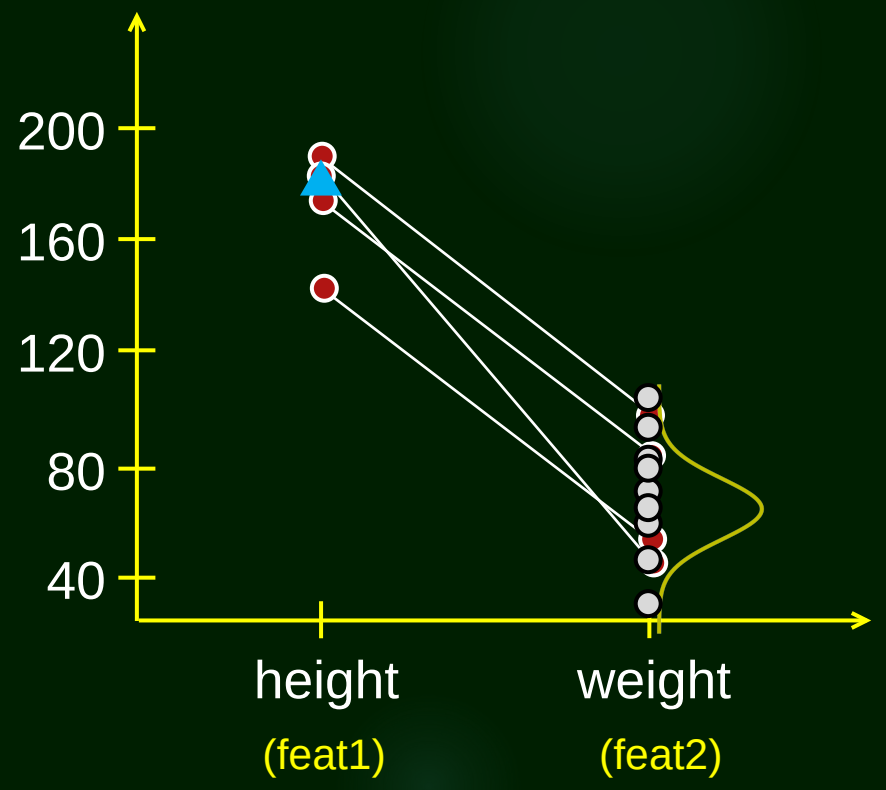
One gets $p(\mathrm{H} \mid \mathrm{W})$ analytically from the original joint distribution $p(\mathrm{W},\mathrm{H})$

Gaussian distribution (2D)

Changing visualization & getting: $p(W \mid H = 180)$



Question: If the height is e.g., 180 cm, what is the most probable weight?



In fact, we don't need to draw many samples to build another gaussian for the weight distribution: we get p(W | H) analytically!

Gaussian distribution

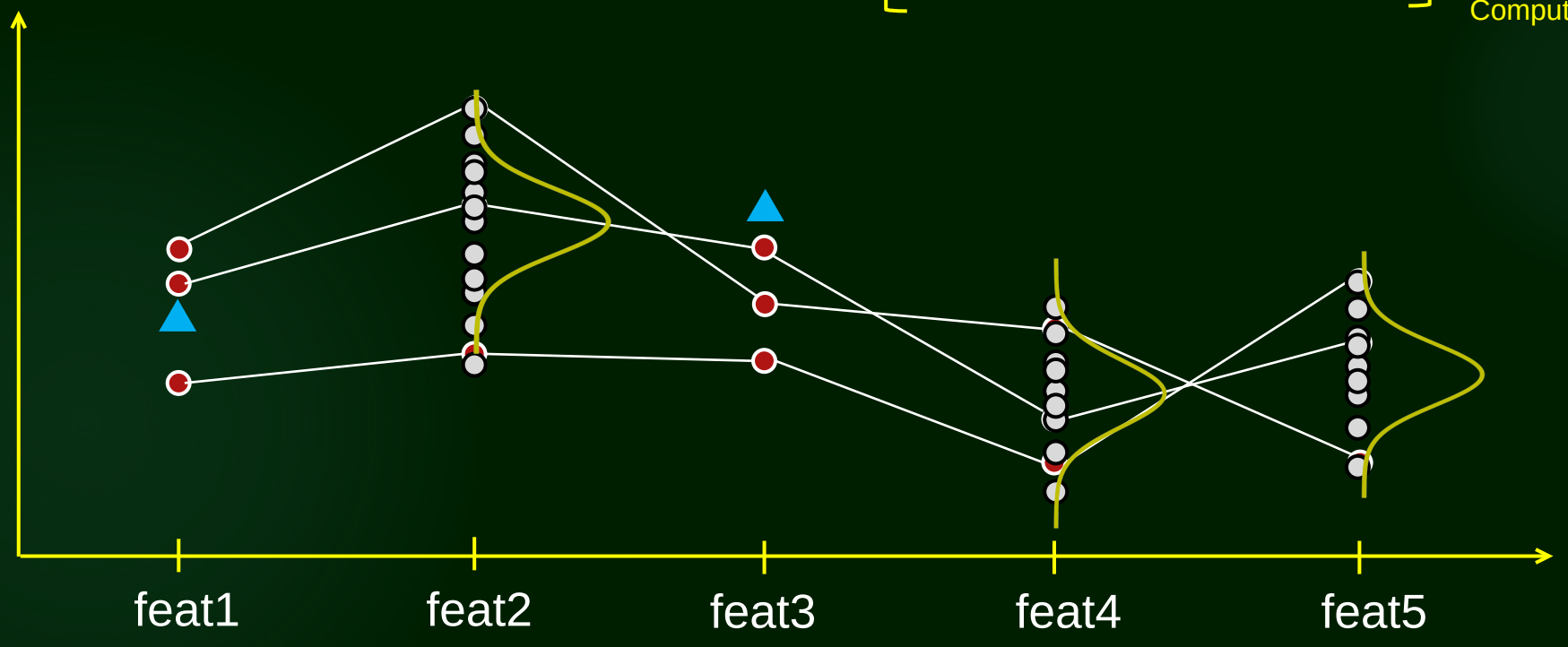
5D Case

Each sample has 5 coordinates, which are different features, named (feat1, feat2, feat3, feat4, feat5)

$$
\begin{array}{c|ccccc}
 & \text{feat1} & \text{feat2} & \text{feat3} & \text{feat4} & \text{feat5} \\
\hline
\text{feat1} & 1 & 0.2 & 0.9 & -0.1 & 0.5 \\
\text{feat2} & 0.2 & 1 & 0.3 & 0.4 & 0.7 \\
\text{feat3} & 0.9 & 0.3 & 1 & -0.6 & 0.5 \\
\text{feat4} & -0.1 & 0.4 & -0.6 & 1 & 0.8 \\
\text{feat5} & 0.5 & 0.7 & 0.5 & 0.8 & 1
\end{array}
$$

Covariance matrix (Σ)
(features standardized)

Computed using observations
(existing data)



Question: If we know feat1, and feat3, what could be feat2, feat4 and feat5?

Towards Gaussian Processes (GP)

1D case

Feature x

Indices:

Different values of the same variable x

Each line:

Drawing from a 5D Gaussian

No measurements done, we define the 5D Gaussian as:

- Mean: $\mu = \vec{0}$
- Covariance: Kernel function

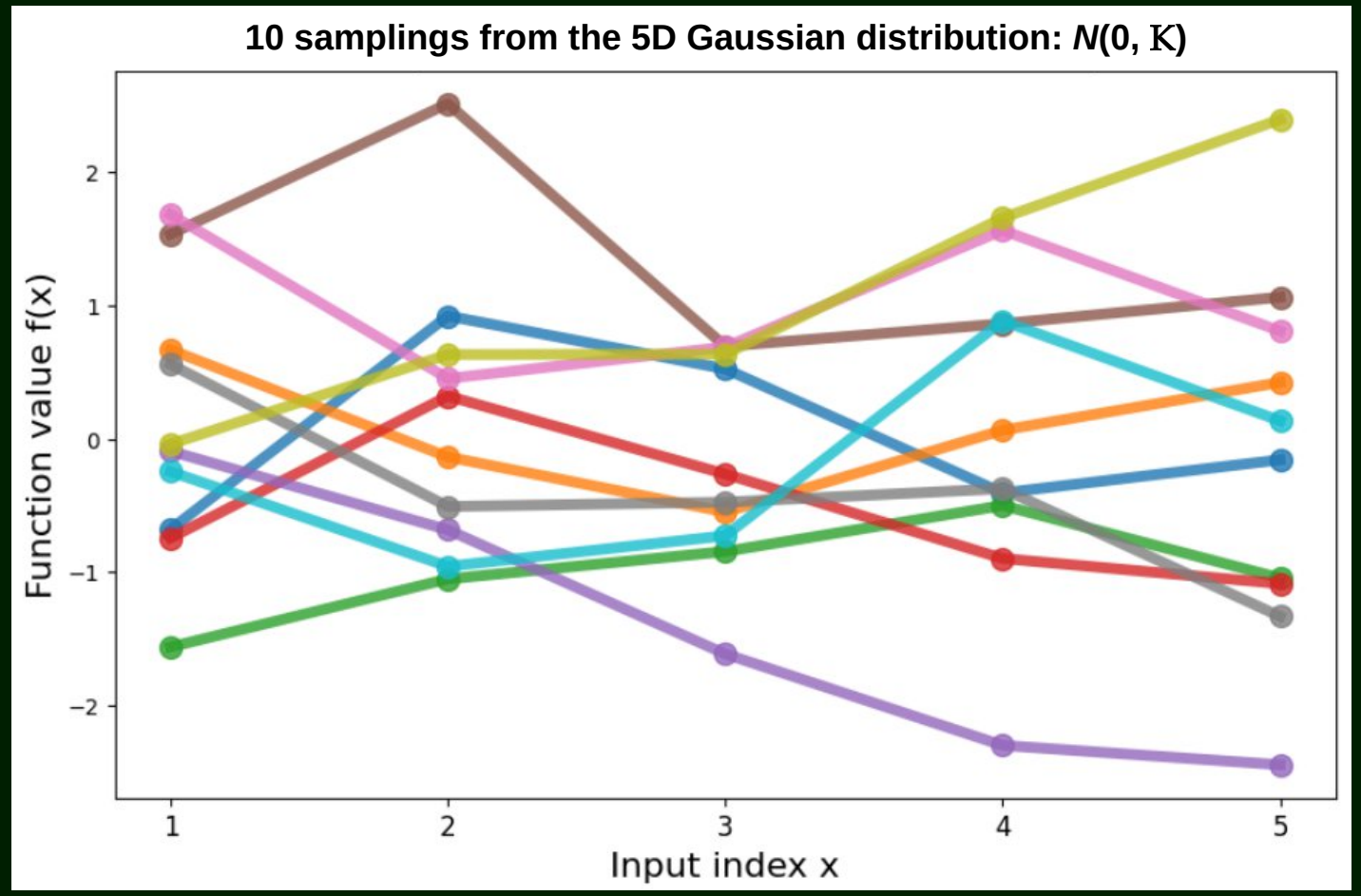
(between samples, not features!)

Prior distribution

$$\mathbf{f}(\mathbf{X}) \sim \mathcal{N}(\mathbf{0}, \mathbf{K}(\mathbf{X},\mathbf{X}))$$ (Noise free!)

(f = underlying or true function)

Conditional distributions can also be extracted!



Each drawing (line) gets 5 numbers, but they are now target values (or f(x)) instead of independent feature values as before. We have a distribution over functions evaluated for different samples (each sample with its own x value)

Towards Gaussian Processes (GP)

Kernels

Similarity between two samples.
E.g., RBF kernel:

$$k(x,x') = \sigma_f^2 \exp\left(-\frac{(x-x')^2}{2\ell^2}\right)$$

ℓ is the length scale (how quickly the correlation decays)
σ_f^2 is the signal variance (the overall vertical scale)

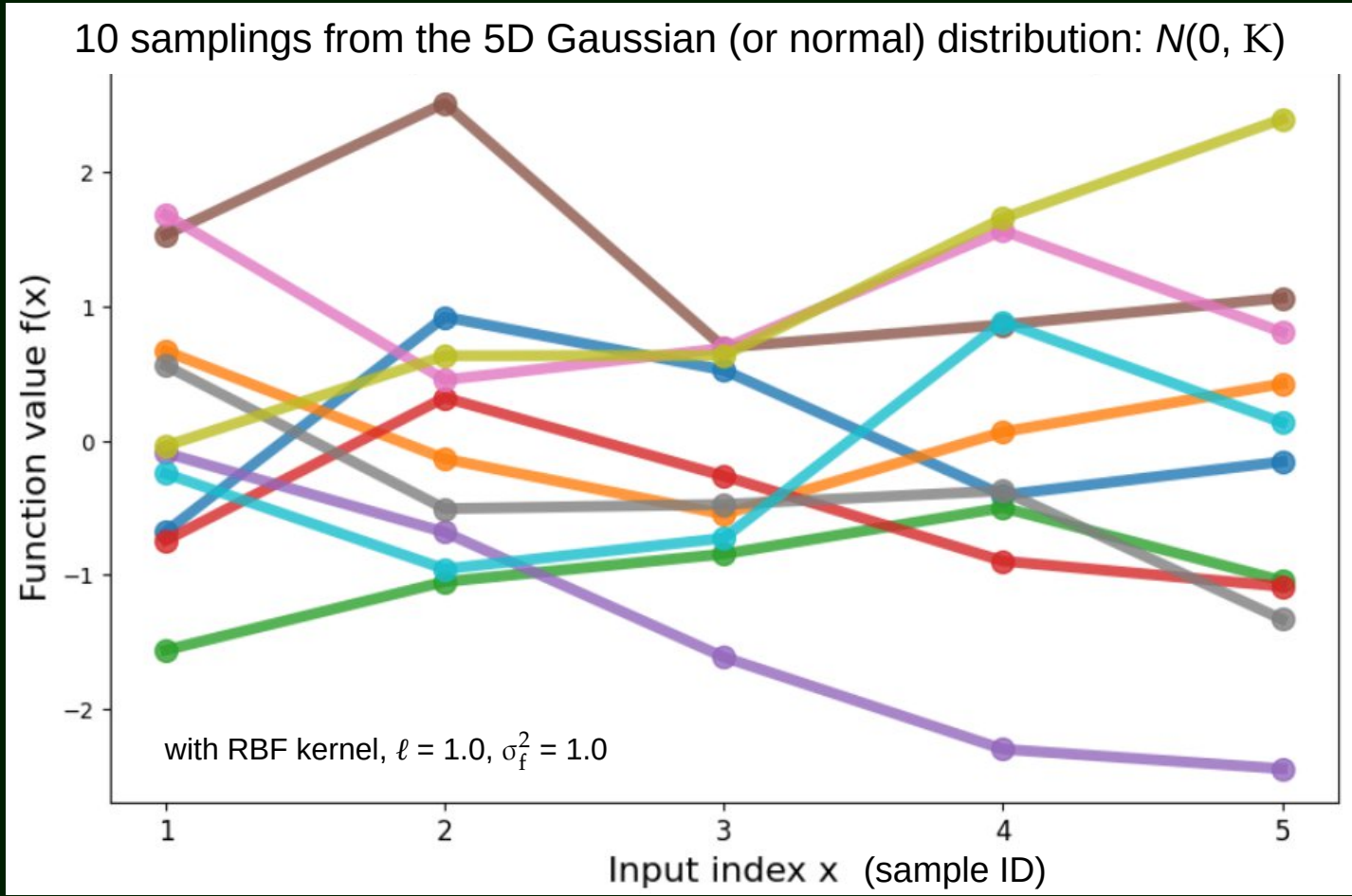
Kernel matrix (K)

samples

samples	1	2	3	4	5
1	1.00	0.61	0.14	0.01	0.00
2	0.61	1.00	0.61	0.14	0.01
3	0.14	0.61	1.00	0.61	0.14
4	0.01	0.14	0.61	1.00	0.61
5	0.00	0.01	0.14	0.61	1.00

Computed using the RBF kernel
with ℓ = 1.0 and σ_f^2 = 1.0
(there are no observations yet!)

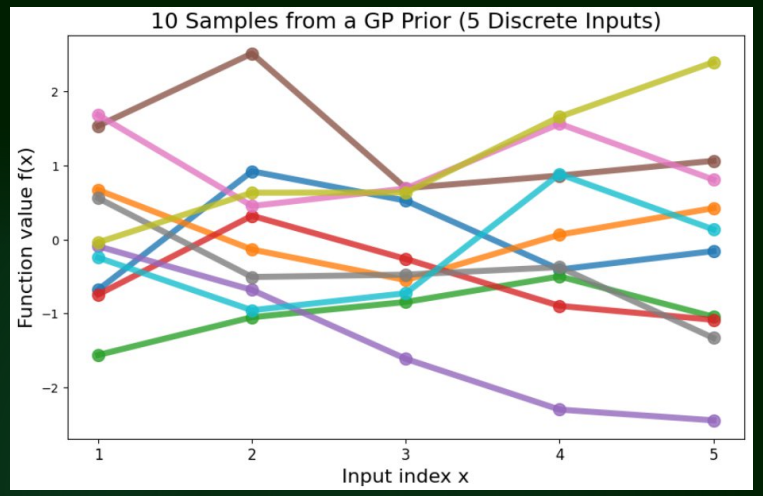
Note: If x1 and x2 are similar (high kernel value), then f(x1) and f(x2) are also similar!



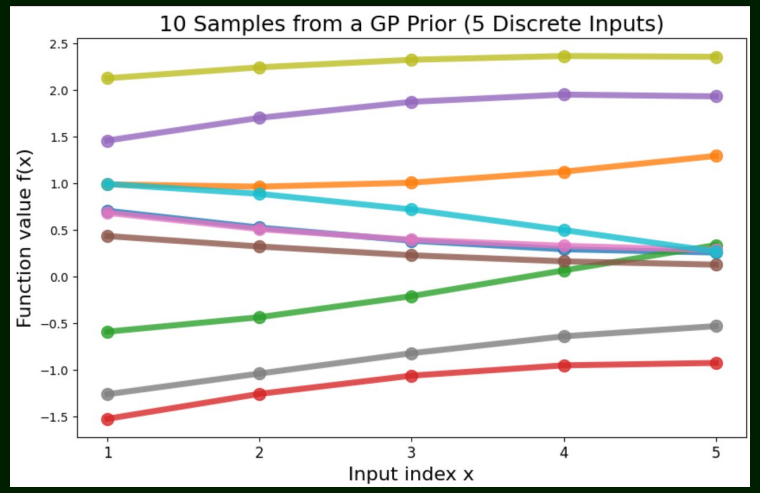
Towards Gaussian Processes (GP)

Influence of the hyperparameters

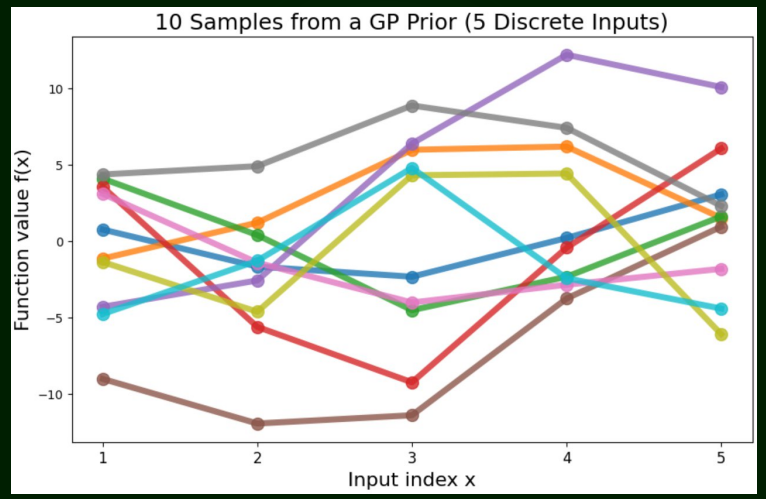
(For the RBF kernel)



ℓ = 1.0, σ_f^2 = 1.0



ℓ = 5.0, σ_f^2 = 1.0



ℓ = 1.0, σ_f^2 = 5.0

In each case, the kernel matrix $\mathbf{K}$ is different

Large ℓ: f(x) is less sensitive to x (feature is less important)

After observations (measurements), these hyperparameters are optimized by taking the log of the likelihood function $p(y \mid X, \theta) \sim N(y \mid 0, K_\theta + \sigma_n^2 I)$, which is maximized via gradient descent.

Gaussian Processes (GP)

1D case

Different values of x

We can have infinite x values. For each x value, one can get also infinite values of f(x) (= samplings from GP)!

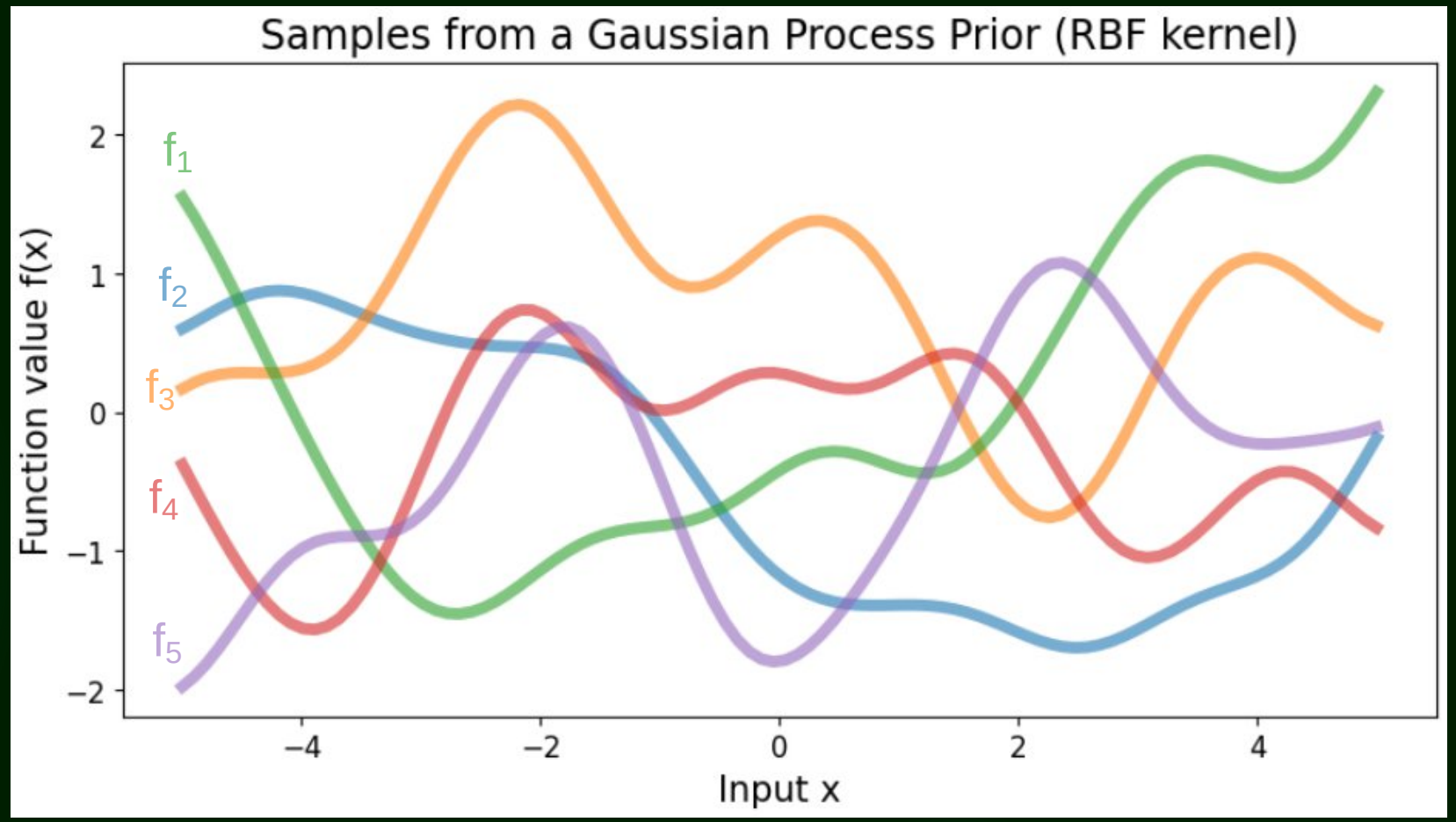
Each line:

Drawing from a 100D Gaussian. No measurements done, we define the 100D Gaussian as:

- Mean: $\mu = \vec{0}$
- Covariance: Kernel function

$$\mathbf{f}(\mathbf{X}) \sim \mathcal{N}(\mathbf{0}, \mathbf{K}(\mathbf{X},\mathbf{X}))$$

Conditional distributions can be extracted



In practice, we don't use infinite values of x, but a small number (e.g., 100, shown above) to facilitate calculations.

Gaussian Processes (GP)

- GP is an infinite-dimensional generalization of a multivariate Gaussian
- It is also a collection of random variables, any finite number of which have a joint Gaussian distribution.
- The elements of the covariance matrix are computed using the kernel function
- The covariance matrix shows similarities between samples, not features

Gaussian Processes (GP)

Prediction of targets (y*) for new sample x*

Measurements (y) have noise: $y_i = f(x_i) + \epsilon_i$ ➡ Gaussian noise: $\epsilon_i \sim \mathcal{N}\left(0, \sigma_n^2\right)$

$$\mathbf{f}(\mathbf{X}) \sim \mathcal{N}(\mathbf{0}, \mathbf{K}(\mathbf{X},\mathbf{X})) \quad \Rightarrow \quad y \sim \mathcal{N}\left(\mathbf{0}, \mathbf{K}(\mathbf{X},\mathbf{X}) + \sigma_n^2\mathbf{I}\right)$$

Noise of the target (hyperparameter)

1) Adopt the prior $y \sim \mathcal{N}\left(\mathbf{0}, \mathbf{K}(\mathbf{X},\mathbf{X}) + \sigma_n^2\mathbf{I}\right)$, built using the training set features (X) and a kernel

2) Assume that y (current target for training set X) and y* (target to be predicted for new point x*) have a joint distribution: $\begin{pmatrix} y \\ y^* \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} \mathbf{K}(\mathbf{X},\mathbf{X}) + \sigma_n^2\mathbf{I} & \mathbf{K}(\mathbf{X},x^*) \\ \mathbf{K}(x^*,\mathbf{X}) & k(x^*,x^*) + \sigma_n^2 \end{bmatrix}\right)$

3) Get the <u>conditional probability</u> $p(y^* \mid \mathrm{X}, y, x^*)$, also called "posterior distribution", from where:

$$\mu^* = \mathbf{K}(x^*,\mathbf{X})\left[\mathbf{K}(\mathbf{X},\mathbf{X}) + \sigma_n^2\mathbf{I}\right]^{-1}\mathbf{y} \qquad \sigma^{*2} = k(x^*,x^*) + \sigma_n^2 - \mathbf{K}(x^*,\mathbf{X})\left[\mathbf{K}(\mathbf{X},\mathbf{X}) + \sigma_n^2\mathbf{I}\right]^{-1}\mathbf{K}(\mathbf{X},x^*)$$

Gaussian Processes (GP)

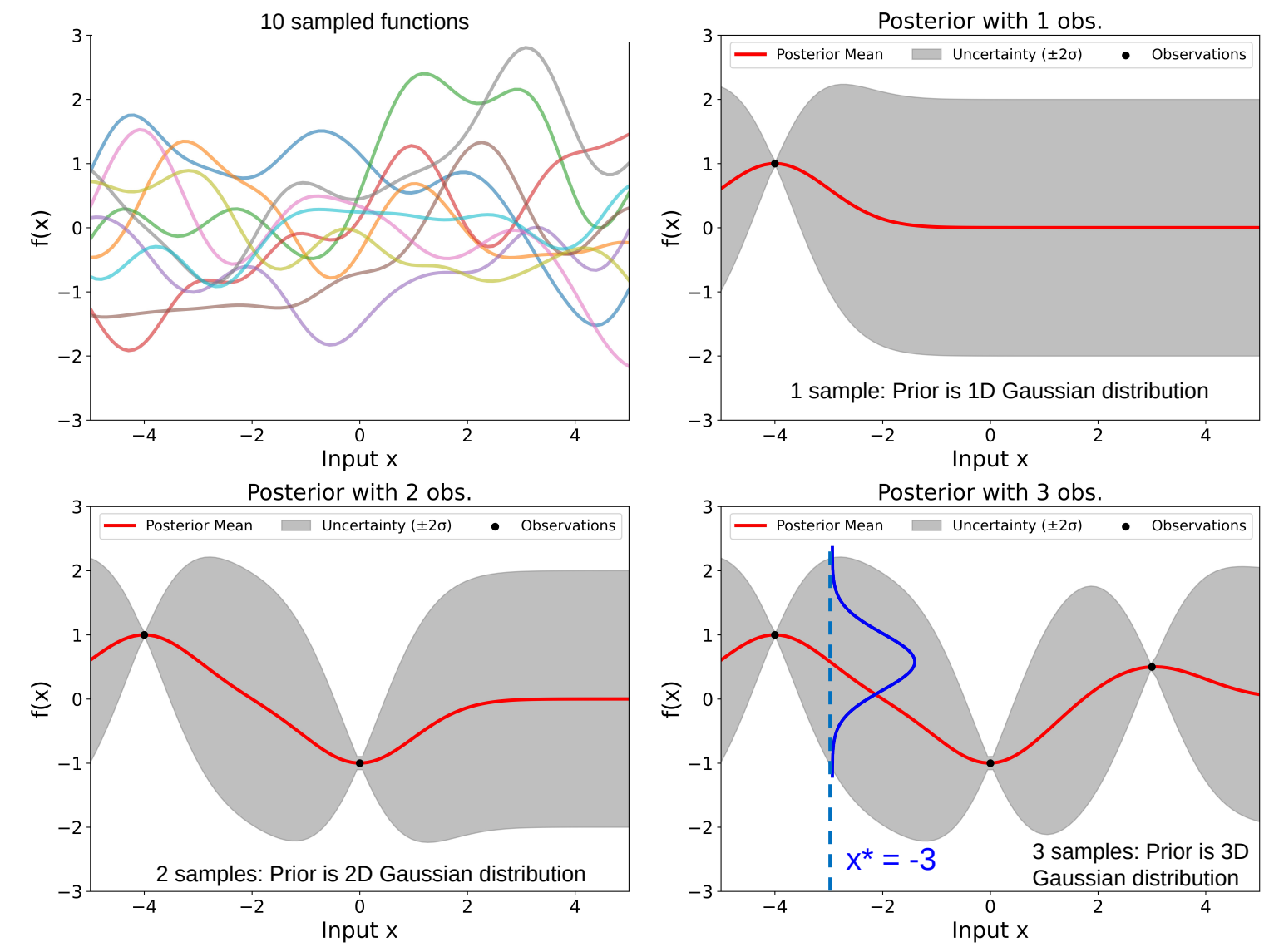
Making observations

1) Upon adding new data:
– Posterior distribution (mean and uncertainty) change!

2) Each prediction is a Gaussian distribution itself (bottom/right blue curve), with mean and variance

$$\mu^* = \mathbf{K}(x^*,\mathbf{X})\left[\mathbf{K}(\mathbf{X},\mathbf{X}) + \sigma_n^2\mathbf{I}\right]^{-1}\mathbf{y}$$

$$\sigma^{*2} = k(x^*,x^*) + \sigma_n^2 - \mathbf{K}(x^*,\mathbf{X})\left[\mathbf{K}(\mathbf{X},\mathbf{X}) + \sigma_n^2\mathbf{I}\right]^{-1}\mathbf{K}(\mathbf{X},x^*)$$



Gaussian Processes (GP)

Multidimensional case

- Euclidean distances $\|\vec{x} - \vec{x}'\|$ can be used instead of $(x - x')$, or:
- One can consider features separately via the ARD (Automatic Relevance Determination) kernel, from where feature importances are computed
- The expressions for the predictive mean and variance remain the same
- The posterior becomes a (hyper)surface (for 1 feature (x), it was a line!)
- The computational cost depends mostly on the number of samples, not features!

Exercises

1) Download the file ‘files9.zip’ from moodle/e-learning (folder “Week 9”) and unzip it locally on your computer
2) Go to google colab (online) **or** launch jupyter notebook locally (offline)
3) Open/upload ‘exercise9.ipynb’ and wait for more instructions